

Creation Date 06-Sep-2010

Revision Date 10-Feb-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	Nitroethane
Cat No. :	L02495
Synonyms	Ethane, nitro
Index No	609-035-00-1
CAS No	79-24-3
Molecular Formula	C ₂ H ₅ N O ₂
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 3 (H226)

Health hazards

Acute oral toxicity

Category 4 (H302)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Warning

Hazard Statements

H226 - Flammable liquid and vapor

H302 + H332 - Harmful if swallowed or if inhaled

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Nitroethane	79-24-3	EEC No. 201-188-9	>95	Flam. Liq. 3 (H226) Acute Tox. 4 (H302) Acute Tox. 4 (H332)

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

REACH registration number

-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Inhalation	Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

Nitrogen oxides (NO_x), Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Nitroethane	TWA: 62 mg/m ³ (8h)	STEL: 100 ppm 15 min	TWA / VME: 100 ppm (8	TWA: 20 ppm 8 uren	STEL / VLA-EC: 100

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

	TWA: 20 ppm (8h) STEL: 312 mg/m ³ (15min) STEL: 100 ppm (15min) Skin	STEL: 312 mg/m ³ 15 min TWA: 20 ppm 8 hr TWA: 62 mg/m ³ 8 hr Skin	heures). TWA / VME: 310 mg/m ³ (8 heures). STEL / VLCT: 100 ppm. indicative limit STEL / VLCT: 312 mg/m ³ . indicative limit Peau	TWA: 62 mg/m ³ 8 uren STEL: 100 ppm 15 minuten STEL: 312 mg/m ³ 15 minuten Huid	ppm (15 minutos). STEL / VLA-EC: 312 mg/m ³ (15 minutos). TWA / VLA-ED: 20 ppm (8 horas) TWA / VLA-ED: 62 mg/m ³ (8 horas) Piel
--	---	--	--	--	--

Component	Italy	Germany	Portugal	The Netherlands	Finland
Nitroethane	TWA: 62 mg/m ³ 8 ore. Time Weighted Average TWA: 20 ppm 8 ore. Time Weighted Average	TWA: 10 ppm (8 Stunden). AGW - exposure factor 4 TWA: 31 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 10 ppm (8 Stunden). MAK TWA: 31 mg/m ³ (8 Stunden). MAK Höhepunkt: 40 ppm Höhepunkt: 124 mg/m ³ Haut	STEL: 312 mg/m ³ 15 minutos STEL: 100 ppm 15 minutos TWA: 20 ppm 8 horas TWA: 62 mg/m ³ 8 horas Pele	huid STEL: 312 mg/m ³ 15 minuten TWA: 62 mg/m ³ 8 uren	TWA: 20 ppm 8 tunteina TWA: 62 mg/m ³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 312 mg/m ³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Nitroethane	Haut MAK-KZGW: 100 ppm 15 Minuten MAK-KZGW: 312 mg/m ³ 15 Minuten MAK-TMW: 20 ppm 8 Stunden MAK-TMW: 62 mg/m ³ 8 Stunden	TWA: 20 ppm 8 timer TWA: 62 mg/m ³ 8 timer STEL: 312 mg/m ³ 15 minutter STEL: 100 ppm 15 minutter Hud	Haut/Peau STEL: 100 ppm 15 Minuten STEL: 312 mg/m ³ 15 Minuten TWA: 20 ppm 8 Stunden TWA: 62 mg/m ³ 8 Stunden	STEL: 186 mg/m ³ 15 minutach TWA: 62 mg/m ³ 8 godzinach	TWA: 20 ppm 8 timer TWA: 62 mg/m ³ 8 timer STEL: 100 ppm 15 minutter. value from the regulation STEL: 312 mg/m ³ 15 minutter. value from the regulation Hud

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Nitroethane	TWA: 62 mg/m ³ TWA: 20 ppm STEL : 312 mg/m ³ STEL : 100 ppm Skin notation	kože TWA-GVI: 20 ppm 8 satima. TWA-GVI: 62 mg/m ³ 8 satima. STEL-KGVI: 100 ppm 15 minutama. STEL-KGVI: 312 mg/m ³ 15 minutama.	TWA: 20 ppm 8 hr. TWA: 62 mg/m ³ 8 hr. STEL: 100 ppm 15 min STEL: 312 mg/m ³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 312 mg/m ³ STEL: 100 ppm TWA: 62 mg/m ³ TWA: 20 ppm	TWA: 60 mg/m ³ 8 hodinách. Ceiling: 300 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Nitroethane	TWA: 20 ppm 8 tundides. TWA: 60 mg/m ³ 8 tundides. STEL: 50 ppm 15 minutites. STEL: 150 mg/m ³ 15 minutites.	Skin notation TWA: 62 mg/m ³ 8 hr TWA: 20 ppm 8 hr STEL: 312 mg/m ³ 15 min STEL: 100 ppm 15 min	skin - potential for cutaneous absorption STEL: 100 ppm STEL: 312 mg/m ³ TWA: 20 ppm TWA: 62 mg/m ³	STEL: 312 mg/m ³ 15 percekben. CK TWA: 62 mg/m ³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 100 ppm STEL: 312 mg/m ³ TWA: 20 ppm 8 klukkustundum. TWA: 62 mg/m ³ 8 klukkustundum. Skin notation

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Nitroethane	skin - potential for cutaneous exposure STEL: 312 mg/m ³ STEL: 100 ppm TWA: 30 mg/m ³ TWA: 9.6 ppm	TWA: 62 mg/m ³ IPRD TWA: 20 ppm IPRD STEL: 150 mg/m ³ STEL: 50 ppm	Possibility of significant uptake through the skin TWA: 20 ppm 8 Stunden TWA: 62 mg/m ³ 8 Stunden STEL: 100 ppm 15 Minuten STEL: 312 mg/m ³ 15 Minuten	possibility of significant uptake through the skin TWA: 20 ppm TWA: 62 mg/m ³ STEL: 100 ppm 15 minuti STEL: 312 mg/m ³ 15 minuti	Skin notation TWA: 20 ppm 8 ore TWA: 62 mg/m ³ 8 ore STEL: 100 ppm 15 minute STEL: 312 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Nitroethane	MAC: 30 mg/m ³	Ceiling: 312 mg/m ³ TWA: 20 ppm	TWA: 20 ppm 8 urah TWA: 62 mg/m ³ 8 urah	Binding STEL: 50 ppm 15 minuter	

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

		TWA: 62 mg/m ³	Koža STEL: 100 ppm 15 minutah STEL: 312 mg/m ³ 15 minutah	Binding STEL: 150 mg/m ³ 15 minuter TLV: 20 ppm 8 timmar. NGV TLV: 62 mg/m ³ 8 timmar. NGV Hud	
--	--	---------------------------	--	--	--

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Nitroethane 79-24-3 (>95)		DNEL = 2100mg/kg bw/day		DNEL = 350mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Nitroethane 79-24-3 (>95)	DNEL = 50mg/m ³	DNEL = 17mg/m ³	DNEL = 25mg/m ³	DNEL = 8.4mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Nitroethane 79-24-3 (>95)	PNEC = 48.8µg/L	PNEC = 0.268mg/kg sediment dw	PNEC = 174µg/L	PNEC = 6mg/L	PNEC = 0.025mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Nitroethane 79-24-3 (>95)	PNEC = 4.88µg/L				

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

Personal protective equipment

Eye Protection Wear safety glasses with side shields (or goggles) (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection No protective equipment is needed under normal use conditions.

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

Small scale/Laboratory use Maintain adequate ventilation

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	Pleasant	
Odor Threshold	No data available	
Melting Point/Range	-90 °C / -130 °F	
Softening Point	No data available	
Boiling Point/Range	112 - 116 °C / 233.6 - 240.8 °F	@ 760 mmHg
Flammability (liquid)	Flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 3.4 Vol%	
Flash Point	28 °C / 82.4 °F	Method - No information available
Autoignition Temperature	414 °C / 777.2 °F	
Decomposition Temperature	No data available	
pH	6.0	0.01M aq.sol
Viscosity	0.677 cP at 20 °C	
Water Solubility	4.6 g/100 mL (20°C)	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Nitroethane	0.18	
Vapor Pressure	21 mbar @ 20 °C	
Density / Specific Gravity	1.040	
Bulk Density	Not applicable	Liquid
Vapor Density	2.59	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C2 H5 N O2
Molecular Weight	75.06
Explosive Properties	explosive air/vapour mixtures possible

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

Yes

10.2. Chemical stability

Hygroscopic.

10.3. Possibility of hazardous reactions

Hazardous Polymerization
Hazardous Reactions

No information available.
Heating may cause an explosion.

10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition. Excess heat.
Incompatible products. Exposure to moist air or water.

10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Aldehydes. Metals. copper. Reducing Agent.

10.6. Hazardous decomposition products

Nitrogen oxides (NO_x). Carbon monoxide (CO). Carbon dioxide (CO₂).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral	Category 4
Dermal	No data available
Inhalation	Category 4

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Nitroethane	LD50 = 1625 mg/kg (Rat)	-	>2200 ppm / 6h (rat)

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory	No data available
Skin	No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Do not empty into drains. .

Component	Freshwater Fish	Water Flea	Freshwater Algae
Nitroethane	LC50: 880 mg/L/48h (Zebrafish)	EC50: 1200 mg/L/24h	

12.2. Persistence and degradability

Persistence Persistence is unlikely.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Nitroethane	0.18	No data available

12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors.

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance.
Ozone Depletion Potential This product does not contain any known or suspected substance.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

Contaminated Packaging	Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2842
14.2. UN proper shipping name	NITROETHANE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

ADR

14.1. UN number	UN2842
14.2. UN proper shipping name	NITROETHANE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

IATA

14.1. UN number	UN2842
14.2. UN proper shipping name	NITROETHANE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Nitroethane	79-24-3	201-188-9	-	-	X	X	KE-25991	X	X

Component	CAS No	TSCA	TSCA Inventory notification -	DSL	NDSL	AICS	NZIoC	PICCS
-----------	--------	------	-------------------------------	-----	------	------	-------	-------

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

			Active-Inactive					
Nitroethane	79-24-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Nitroethane	79-24-3	-	-	-

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Nitroethane	79-24-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Nitroethane	WGK2	Class II : 0.10 g/m ³ (Massenkonzentration)

Component	France - INRS (Tables of occupational diseases)
Nitroethane	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H226 - Flammable liquid and vapor

H302 - Harmful if swallowed

SAFETY DATA SHEET

Nitroethane

Revision Date 10-Feb-2024

H332 - Harmful if inhaled

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Prepared By

Health, Safety and Environmental Department

Creation Date

06-Sep-2010

Revision Date

10-Feb-2024

Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet